

EXPERIMENT-11

Query with Joins – EquiJoin, InnerJoin, OuterJoin

```
mysql> create database collegedb;
Query OK, 1 row affected (0.06 sec)

mysql> use collegedb;
Database changed
mysql> create table department(dept_id int primary key,dept_name varchar(30));
Query OK, 0 rows affected (0.10 sec)

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| dept_id | int | NO | PRI | NULL | |
| dept_name | varchar(30) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> create table faculty(faculty_id int primary key,faculty_name varchar(30),dept_id int,foreign key(dept_id) references department(dept_id));
Query OK, 0 rows affected (0.10 sec)

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| faculty_id | int | NO | PRI | NULL | |
| faculty_name | varchar(30) | YES | | NULL | |
| dept_id | int | YES | MUL | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> insert into department values(1,'computer science'),(2,'electronics'),(3,'mechanical'),(4,'civil');
Query OK, 4 rows affected (0.15 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> select*from department;
+-----+-----+
| dept_id | dept_name |
+-----+-----+
| 1 | computer science |
| 2 | electronics |
| 3 | mechanical |
| 4 | civil |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> insert into faculty values(101,'sonu',1),(102,'pranathi',2),(103,'chownu',3),(104,'roops',NULL);
Query OK, 4 rows affected (0.02 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> select*from faculty;
+-----+-----+
| faculty_id | faculty_name | dept_id |
+-----+-----+
| 101 | sonu | 1 |
| 102 | pranathi | 2 |
| 103 | chownu | 3 |
| 104 | roops | NULL |
+-----+-----+
4 rows in set (0.06 sec)
```

```
mysql> select faculty_name,dept_name
-> from faculty,department
-> where faculty.dept_id=department.dept_id;
+-----+-----+
| faculty_name | dept_name |
+-----+-----+
| sonu | computer science |
| pranathi | electronics |
| chownu | mechanical |
+-----+-----+
3 rows in set (0.07 sec)
```

```
mysql> select faculty_name,dept_name
-> from faculty
-> inner join department
-> on faculty.dept_id=department.dept_id;
+-----+-----+
| faculty_name | dept_name |
+-----+-----+
| sonu | computer science |
| pranathi | electronics |
| chownu | mechanical |
+-----+-----+
3 rows in set (0.01 sec)
```

```
mysql> select faculty_name,dept_name
-> from faculty
-> left outer join department
-> on faculty.dept_id=department.dept_id;
+-----+-----+
| faculty_name | dept_name |
+-----+-----+
```

```
| faculty_name | dept_name |
+-----+-----+
| sonu | computer science |
| pranathi | electronics |
| chownu | mechanical |
+-----+-----+
3 rows in set (0.01 sec)
```

```
mysql> select faculty_name,dept_name
-> from faculty
-> left outer join department
-> on faculty.dept_id=department.dept_id;
+-----+-----+
| faculty_name | dept_name |
+-----+-----+
| sonu | computer science |
| pranathi | electronics |
| chownu | mechanical |
| roops | NULL |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> select faculty_name,dept_name
-> from faculty
-> right outer join department
-> on faculty.dept_id=department.dept_id;
+-----+-----+
| faculty_name | dept_name |
+-----+-----+
| sonu | computer science |
| pranathi | electronics |
| chownu | mechanical |
| NULL | civil |
+-----+-----+
4 rows in set (0.00 sec)
```